

9-1 · Explore Relationships Between Two Variables

Connect & apply · 8 tasks · Prepares for 6.EE.C.9

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Evaluate $4x + 1$ when $x = 3$.

13

Substitute 3: $4 \times 3 + 1 = 13$.

Skill: Evaluate an expression for several input values

2. FOUNDATION 2

For $y = 2x + 1$, give y for $x = 0, 1, 2$ in that order.

1, 3, 5

Substitute each input; the output grows by 2 per step.

Skill: Build a table of values from a rule

3. FOUNDATION 3

Tickets cost \$6 each. What is the cost in dollars of 7 tickets?

42

Multiply the unit rate by the quantity: 6×7 .

Skill: Multiply a unit rate by a quantity

4. FOUNDATION 4

You choose how many notebooks to buy. The total cost changes with that number. Which quantity is independent?

Number of notebooks

It is the input you choose; total cost depends on it.

Skill: Tell which quantity is chosen and which one responds

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5. CHECK A

Songs cost \$2 each. Which quantity is independent? Complete the table for 1, 2, and 3 songs.

Independent variable: Number of songs purchased (x). Dependent variable: Total cost (y). Table of values: (1 song, \$2), (2 songs, \$4), (3 songs, \$6).

Compare each part of your work with the worked solution.

6. CHECK B

Stickers cost \$4 each. Which quantity is independent? Complete the table for 1, 2, and 3 stickers.

Independent: number of stickers (x). Dependent: total cost (y). Table: (1, \$4), (2, \$8), (3, \$12).

Compare each part of your work with the worked solution.

7. CHECK C

A plant grows 5 cm per week. Which quantity is independent? Complete the table for 1, 2, and 3 weeks.

Independent: number of weeks (x). Dependent: height gained (y). Table: (1, 5), (2, 10), (3, 15).

Compare each part of your work with the worked solution.

8. TRANSFER

A gym charges a \$20 join fee plus \$15 per month. Is the total still a simple per-month rate? Explain.

No. The table does not start at (0, 0) — at 0 months the cost is \$20. The rule is $y = 15x + 20$, not $y = 15x$.

Explain why your method works and include units when needed.

9-2 · Analyze Graphs of Relationships Between Two Variables

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Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

A table has $x = 3$ and $y = 12$ in one row. Write its ordered pair.

(3, 12)

Pair the x and y from the same row, with x first.

Skill: Plot ordered pairs taken from a table

2. FOUNDATION 2

A graph point is 5 units right and 20 units up. Give its coordinates.

(5, 20)

Coordinates report units, not a count of drawn tick marks.

Skill: Read a point's coordinates off a graph

3. FOUNDATION 3

A y -axis is labeled 0, 10, 20, 30. What value lies halfway between 10 and 20?

15

Half of a 10-unit interval is 5; $10 + 5 = 15$.

Skill: Read the scale on each axis

4. FOUNDATION 4

Continue the points (1, 4), (2, 8), (3, 12) using $y = 4x$. Give the point for $x = 4$.

(4, 16)

Substitute $x = 4$ in the rule: $y = 16$.

Skill: Extend a pattern to predict the next point

9-2 · Analyze Graphs of Relationships Between Two Variables

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5. CHECK A

In (x, y) , which value comes first? Read the point at 4 across and 12 up.

In (x, y) , the x -value (horizontal coordinate) always comes first, followed by y (vertical coordinate). Point coordinates: $(4, 12)$.

Compare each part of your work with the worked solution.

6. CHECK B

In (x, y) , which value comes first? Read the point at 3 across and 15 up.

The x -value comes first. The point is $(3, 15)$.

Compare each part of your work with the worked solution.

7. CHECK C

In (x, y) , which axis is the independent variable on? Read the point at 6 across and 24 up.

The independent variable is on the horizontal x -axis. The point is $(6, 24)$.

Compare each part of your work with the worked solution.

8. TRANSFER

A graph of a ratio relationship always passes through which point, and why?

$(0, 0)$. When the independent quantity is zero, the dependent quantity is zero too.

Explain why your method works and include units when needed.

9-3 · Write Equations to Represent Relationships Between Two Variables

Connect & apply · 8 tasks · Prepares for 6.EE.C.9

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

A proportional table contains (2, 10), (4, 20), (6, 30). Find the unit rate y/x .

5

Each quotient is 5: $10 \div 2$, $20 \div 4$, and $30 \div 6$.

Skill: Find the unit rate from a table

2. FOUNDATION 2

For $x = 8$ and $y = 28$, compute y/x .

3.5

$28 \div 8 = 3.5$.

Skill: Divide y by x to find the constant factor

3. FOUNDATION 3

Write an expression for the cost of n notebooks at \$4 each.

$4n$

Multiply dollars per notebook by the number of notebooks.

Skill: Write an expression using a variable

4. FOUNDATION 4

Does $y = 3x$ fit every pair (1, 3), (2, 6), (3, 10)? Explain.

No; (3, 10) fails

At $x = 3$ the equation predicts 9, not 10; every row must fit.

Skill: Test an equation against every row of the table

9-3 · Write Equations to Represent Relationships Between Two Variables

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5. CHECK A

Table: (1, 3), (2, 6), (3, 9). Write y in terms of x .

Rule / Equation: $y = 3x$ (or $y = 3 \cdot x$). Unit rate is 3 ($y \div x = 3 \div 1 = 6 \div 2 = 9 \div 3 = 3$).

Compare each part of your work with the worked solution.

6. CHECK B

Table: (1, 5), (2, 10), (3, 15). Write y in terms of x .

$y = 5x$. Every y divided by its x gives 5.

Compare each part of your work with the worked solution.

7. CHECK C

Table: (2, 14), (4, 28), (6, 42). Write y in terms of x .

$y = 7x$. $14 \div 2 = 7$, $28 \div 4 = 7$, $42 \div 6 = 7$.

Compare each part of your work with the worked solution.

8. TRANSFER

A table gives (3, 12), (5, 20), (8, 32). Write the equation and predict y when $x = 11$.

$y \div x = 4$ for every row, so $y = 4x$. At $x = 11$, $y = 44$.

Explain why your method works and include units when needed.

9-4 · Apply Two-Variable Relationships to Solve Problems

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Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

For $C = 2.5n$, find C when $n = 6$.

15

Substitute and multiply: $2.5 \times 6 = 15$.

Skill: Substitute into an equation and evaluate

2. FOUNDATION 2

Solve $4x = 28$.

7

Divide both sides by 4.

Skill: Solve a one-step equation for the other variable

3. FOUNDATION 3

Compute $18.9 \div 3$.

6.3

189 tenths divided by 3 is 63 tenths.

Skill: Multiply and divide with decimals

4. FOUNDATION 4

A runner travels at 8 miles per hour for 3 hours.
State the distance with its unit.

24 miles

Multiply rate by time: $8 \times 3 = 24$ miles. The result is distance, not speed.

Skill: Read the answer back in the situation's units

9-4 · Apply Two-Variable Relationships to Solve Problems

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5. CHECK A

Muffins cost \$3 each, so $y = 3x$. How many muffins can you buy with \$24?

Substitute $y = 24$ into $y = 3x$: $24 = 3x$. Divide both sides by 3: $x = 24 \div 3 = 8$ muffins. You can buy 8 muffins.

Compare each part of your work with the worked solution.

6. CHECK B

Bagels cost \$2 each, so $y = 2x$. How many bagels can you buy with \$18?

$18 = 2x$, so $x = 9$ bagels.

Compare each part of your work with the worked solution.

7. CHECK C

A printer makes 15 pages per minute, so $y = 15x$. How long to print 90 pages?

$90 = 15x$, so $x = 6$ minutes.

Compare each part of your work with the worked solution.

8. TRANSFER

A taxi charges \$2.50 per mile. Write the equation and find how far \$20 takes you. Round to the nearest tenth.

$y = 2.5x$. Then $20 = 2.5x$, so $x = 8$ miles exactly.

Explain why your method works and include units when needed.